

Broadcast/broadband integration. Broadband connections offer high-speed data transmission for various online services. In contrast, broadcast systems efficiently deliver over-the-air (OTA) television signals to large audiences simultaneously. D2M technology integrates broadband and broadcast networks to optimise content delivery. By shifting bandwidth-heavy applications, such as live streaming and OTT services, to broadcast channels, D2M reduces strain on cellular networks and enhances overall performance.

Reliable reception. D2M uses advanced error correction techniques and robust transmission methods



D2M is similar to direct-to-home (DTH) technology (Source: ChatGPT)

to provide reliable reception, even in areas where signals are weak or mobility is high.

Targeted advertising. D2M can reach millions of 'TV dark' homes and supports multiple audio tracks, subtitles, and languages, enabling targeted advertising tailored to user location, preferences, and demographics.

Integration with emergency alert systems. D2M can be integrated with emergency alert broadcast systems to deliver critical information and public safety alerts directly to millions of mobile device users, even in areas where internet connectivity is unavailable or disrupted.

Content protection and security. The technology ensures content protection, digital rights management (DRM), intellectual property rights (IPR) safeguards, increased security, protection against cybercrimes, and prevention of unauthorised access.

Energy efficiency. D2M is energy-efficient, minimising power consumption across broadcasting infrastructure and mobile devices.

Enhancing user experience. D2M technology enhances user experiences across multiple environments, such as railways, airports, shopping malls, moving vehicles, and inflight settings.

It provides timely updates on schedules, routes, and safety, alongside seamless entertainment options. Passengers can receive updates on train schedules, delays, platform changes, and entertainment options such as movies, music, and games directly on their mobiles. D2M broadcasts can provide flight updates, gate changes, boarding

times, and interactive maps to help passengers navigate areas such as gates and lounges. D2M broadcast will enable inflight passengers to stream movies, TV shows, music, games and receive updates on the flight status, destination and connecting flight without using Wi-Fi connection. Stores in shopping malls can use D2M broadcast to provide information to shoppers about their promotion offers, discounts, new products and even guide them on locating stores, restrooms, and other amenities.

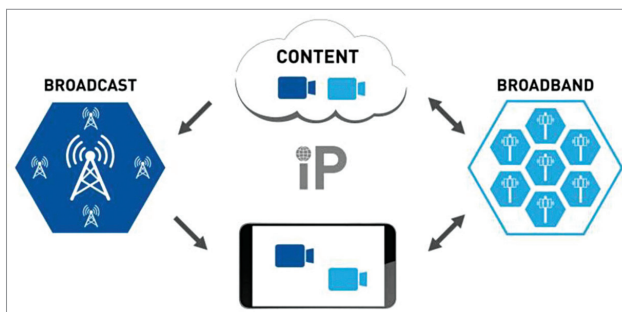
Scalability and future-readiness. The technology is scalable and adaptable to advanced technologies and rapidly evolving mobile devices and networks.

How D2M technology is likely to work in India

The foundation laid down by IIT Kanpur. A conclave, 'D2M Broadcast & 5G Broadband - Convergence Roadmap for India,' was organised by IIT Kanpur in New Delhi in June 2022 to provide insights. A white paper highlighted the immense potential of the D2M project as a viable solution for a substantial user base that either lacks access to digital streaming or suffers from poor internet connectivity.

The conclave, endorsed by various policymakers, industry representatives, and stakeholders, emphasised the role of D2M in optimising video transmission and content consumption in India. It also called for policies and regulations to be framed by the government and regulators with the involvement of all stakeholders.

With an ever-expanding smartphone base and explosive growth in video consumption, the existing wireless infrastructure has struggled to meet the rising demand for content delivery to millions of end users. The D2M broadcast project has the potential to address this demand without relying on costly and often unreliable mobile broadband networks.



D2M technology enables convergence of the broadband and broadcast networks (Source: IIT Kanpur, June 2022 white paper)



The topmost advantages of D2M technology (Source: <https://www.dreamcast.in>)